

Fair Test Practice

Level 1 · Ages 7-9 · A fair test changes just ONE thing. Scientists also repeat tests to be sure.

Name _____ Date _____

- ① Maya drops the same ball from the same height onto grass, and then onto concrete. She measures how high the ball bounces each time.

What changed? _____

What was measured? _____

One thing kept the same: _____

- ② Leo puts one ice cube on a plate in a sunny window and one on the same kind of plate in a shady room. The ice cubes are the same size. He times how long each one takes to melt.

What changed? _____

What was measured? _____

One thing kept the same: _____

- ③ Sam grows three bean plants in the same kind of pot on the same windowsill. Every day one plant gets one spoonful of water, one gets two and one gets three. After two weeks he measures how tall each plant is.

What changed? _____

What was measured? _____

One thing kept the same: _____

- ④ Ella puts one spoonful of sugar in a glass of cold water and one spoonful in a glass of warm water. Both glasses hold the same amount of water. She stirs each one at the same speed until she cannot see any sugar, and times how long it takes.

What changed? _____

What was measured? _____

One thing kept the same: _____

- ⑤ Ava folds two paper planes from the same kind of paper. One has short wings and one has long wings. She throws each plane the same way and measures how far it flies.

What changed? _____

What was measured? _____

One thing kept the same: _____

Identifying Variables — Set A

Level 2 · Ages 9-11 · Read each experiment. Name the independent and dependent variables, and two controlled variables.

Name _____ Date _____

Independent variable — the one thing you change on purpose.

Dependent variable — what you measure to see what happened.

Controlled variables — things you keep the same so the test is fair.

- ① Priya lets a toy car roll down a ramp set at heights of 10 cm, 20 cm and 30 cm. She measures how far the car rolls across the floor. She uses the same car and the same ramp every time.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

- ② Jack ties a metal washer to a string and lets it swing. He tries strings 20 cm, 40 cm and 60 cm long with the same washer, pulling it back the same distance each time. He counts the swings in 30 seconds.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

- ③ An adult pours 200 mL of warm water at 45 °C into each of three identical cups. Omar wraps one cup in foil, one in wool and one in newspaper. After 20 minutes he measures the temperature of the water in each cup.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

Identifying Variables — Set B

Level 2 · Ages 9-11 · Read each experiment. Name the independent and dependent variables, and two controlled variables.

Name _____ Date _____

Independent variable — the one thing you change on purpose.

Dependent variable — what you measure to see what happened.

Controlled variables — things you keep the same so the test is fair.

- ① Lucy puts 1, 2, 3 and then 4 sheets of paper between a magnet and a pile of paper clips. Each time she counts how many clips the magnet picks up. She uses the same magnet and the same paper.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

- ② Ben adds one teaspoon of sugar to 100 mL of water at 20 °C, 30 °C and 40 °C, stirring steadily the whole time. He times how long the sugar takes to dissolve.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

- ③ Zara puts three identical bean plants in a bright window, a shady corner and a dark cupboard. Each plant gets 50 mL of water every day. After two weeks she counts the leaves on each plant.

Independent variable: _____

Dependent variable: _____

Two controlled variables: _____

Identifying Variables — Set A

Level 3 · Ages 11-14 · Name every variable. Give values and units where you can.

Name _____ Date _____

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① Twenty bean seedlings grow in identical pots of the same potting soil on one windowsill, each given 100 mL of water a day. Each week, groups of five get 0, 5, 10 or 15 drops of liquid fertilizer. After four weeks, every plant's height is measured.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ② Students take their pulse for 15 seconds after 2 minutes each of sitting, walking and jogging, resting until it is back to the sitting value in between. Each count $\times 4$ gives beats per minute. (Anyone who should not jog can record.)

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ③ Twenty radish seeds are placed on damp paper towel in each of four identical dishes with lids, kept at 5 °C, 15 °C, 25 °C and 35 °C. Each dish gets 10 mL of water. After five days the seeds that have sprouted are counted.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

Identifying Variables — Set B

Level 3 · Ages 11-14 · Name every variable. Give values and units where you can.

Name _____ Date _____

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① A student drops one fizzing vitamin tablet into 200 mL of water at 10 °C, 25 °C and 40 °C, using a new tablet of the same kind each time and never stirring. She times how long each takes to stop fizzing. (Do not drink the solution.)

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ② Parachutes made from the same paper, with canopy areas of 100, 200, 300 and 400 cm², each carry the same paper-clip load. Each is dropped indoors from 2 m, three times, and the mean time to reach the floor is recorded.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

- ③ A piece of pondweed in water is lit by a lamp placed 10, 20, 30 and 40 cm away. The water is kept at 25 °C. At each distance the gas bubbles (which contain oxygen) given off in one minute are counted.

Independent variable (and values): _____

Dependent variable (and units): _____

Three controlled variables: _____

Control group: _____

Spot the Unfair Test — Set A

Level 4 · Ages 10-13 · Each of these tests has a problem. Find it, then explain how to fix it.

Name _____ Date _____

- ① Kai wants to know which of two kinds of paper towel soaks up more water. He uses a whole sheet of kind A and half a sheet of kind B, dips each in water for 10 seconds, then squeezes out the water and measures it.

What is the test trying to find out?

What is wrong with it? _____

How would you fix it? _____

- ② Mia wants to see if music helps plants grow. She puts one plant with music playing in her sunny bedroom, and one plant with no music in a dark kitchen.

What is the test trying to find out?

What is wrong with it? _____

How would you fix it? _____

- ③ Tom wants to see if the height a ball is dropped from changes how high it bounces. He drops a tennis ball from 50 cm and a basketball from 100 cm.

What is the test trying to find out?

What is wrong with it? _____

How would you fix it? _____

Spot the Unfair Test — Set B

Level 4 · Ages 10-13 · Each of these tests has a problem. Find it, then explain how to fix it.

Name _____ Date _____

- ① Aisha wants to find out whether sugar or salt dissolves faster. She stirs a spoonful of sugar into hot water and a spoonful of salt into cold water, and times each one.

What is the test trying to find out?

What is wrong with it? _____

How would you fix it? _____

- ② Leo wants to know if heavier toy cars roll farther. He rolls a light car and a heavy car down the same ramp once each and measures how far each one goes.

What is the test trying to find out?

What is wrong with it? _____

How would you fix it? _____

- ③ A class gives a new plant food to 10 bean plants. After three weeks the plants have grown 12 cm on average, so the class decides that the plant food works.

What is the test trying to find out?

What is wrong with it? _____

How would you fix it? _____

Types of Variables

Ages 10-14 · Independent, dependent and controlled variables — and how a control group is different.

Name _____ Date _____

Independent variable (I) — the one factor you change on purpose.

Dependent variable (D) — the factor you measure to see the effect.

Controlled variable (C) — a factor you keep the same in every trial, so the test is fair.

Control group (G) — a separate group that gets no treatment, used as a baseline to compare against.

Part A — Sort the factors

Experiment: Does plant food make bean plants grow taller? Four groups of bean plants get 0 g, 2 g, 4 g or 6 g of plant food each week. Everything else is kept the same for all the plants. After three weeks, every plant's height is measured.

Write I, D, C or G in each box.

- ☐ 1. The amount of plant food each week
- ☐ 2. The height of the plants after three weeks
- ☐ 3. The amount of water each plant gets
- ☐ 4. The type of bean plant
- ☐ 5. The group given 0 g of plant food
- ☐ 6. The size of the pots
- ☐ 7. How much light the plants get
- ☐ 8. How long the test lasts (three weeks)

Part B — Find the variables in each hypothesis

1. If the ramp is higher, then the toy car will roll farther.

Independent: _____ **Dependent:** _____

2. If a plant gets more hours of light each day, then it will grow more leaves.

Independent: _____ **Dependent:** _____

3. The warmer the water, the faster salt will dissolve in it.

Independent: _____ **Dependent:** _____

4. If you exercise for longer, then your heart rate will be higher.

Independent: _____ **Dependent:** _____

Part C — Controlled variable or control group?

Circle the right answer for each statement.

1. Every pot gets exactly the same amount of water.

Controlled variable Control group

2. Some plants get no plant food, so the others can be compared with them.

Controlled variable Control group

3. All the plants are kept in the same room.

Controlled variable Control group

Variables in Tables and Graphs

Ages 11-14 · The independent variable goes in the first column of a table and on the x-axis of a graph. The dependent variable goes on the y-axis.

Name _____ Date _____

Example data, made up for practice.

Table 1 (example data)

Ramp height (cm)	Distance rolled (cm)
10	45
20	88
30	131
40	170

Independent variable: _____

Dependent variable: _____

Pattern: _____

Table 2 (example data)

Water temperature (°C)	Time to dissolve (s)
20	95
30	70
40	52
50	38

Independent variable: _____

Dependent variable: _____

Pattern: _____

Table 3 (example data)

Lamp distance (cm)	Bubbles per minute
10	48
20	30
30	18
40	11

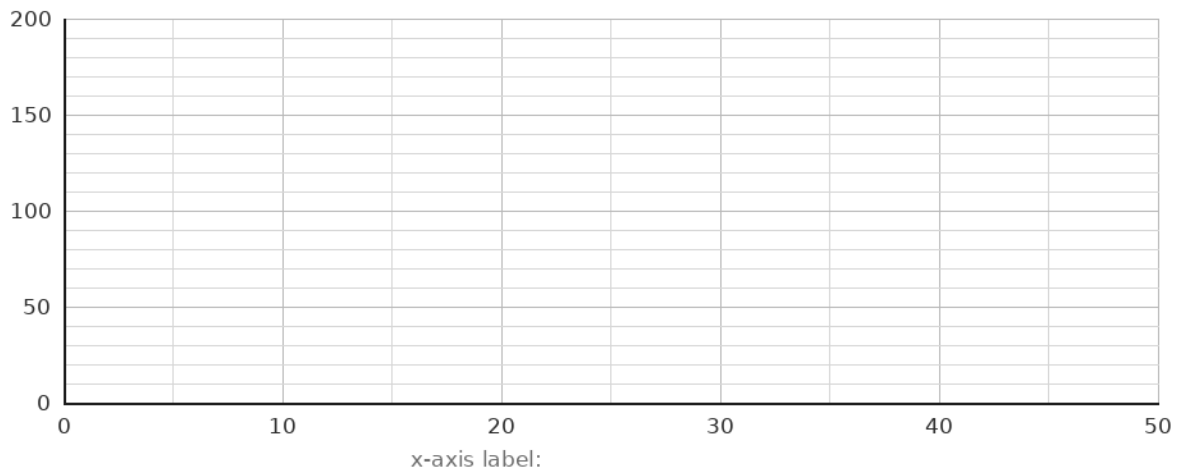
Independent variable: _____

Dependent variable: _____

Pattern: _____

Plot Table 1 on the graph and label both axes.

y-axis label: _____



Making a Fair Test

A fair test changes only one thing. Everything else stays the same.

Name _____ Date _____

Worked example

Fair test: Does a paper plane fly farther with a paper clip on it?

Change: the paper clip. Measure: how far it flies.

Keep the same: the same plane, the same throw, the same starting spot.

My investigation question

The one thing I change I only change this.

The thing I watch or measure This is what I find out.

The things I keep the same List everything that must not change.

Variables in an Experiment

Change one variable, measure one variable, control the rest. That is what makes a test fair.

Name _____ Date _____

Worked example

Question: Does ramp height change how far a toy car rolls?

Independent: ramp height (cm).

Dependent: distance rolled (cm), measured with a ruler.

Controlled: same car, same ramp surface, same starting line.

My investigation question

Independent variable The one thing you change on purpose.

Dependent variable The thing you measure. Give the units.

Controlled variables Everything you deliberately keep the same.

Is my test fair? If someone changed something else as well, could you still trust the result?

Variables and Controls

Identify every variable before you start. Uncontrolled variables are the usual reason a result cannot be trusted.

Name _____ Date _____

Worked example

Question: How does salt concentration affect the melting time of ice?

Independent: salt concentration (0, 5, 10, 15 g per 100 ml).

Dependent: time to melt (s), measured with a stopwatch.

Controlled: ice mass, starting temperature, container, room temperature.

Control group: ice in pure water (0 g of salt).

My investigation question

Independent variable What you change, and the range of values you will use.

Dependent variable What you measure, with units and the instrument used.

Controlled variables What you hold constant, and how you will hold it constant.

Is my test fair? If someone changed something else as well, could you still trust the result?

Fair Test Practice — Answer Key

Level 1 · Ages 7-9 · Answers in bold. Accept any answer that means the same.

- ① Maya drops the same ball from the same height onto grass, and then onto concrete. She measures how high the ball bounces each time.

What changed? The surface the ball lands on (grass or concrete).

What was measured? How high the ball bounces.

One thing kept the same: The ball, or the height she drops it from.

- ② Leo puts one ice cube on a plate in a sunny window and one on the same kind of plate in a shady room. The ice cubes are the same size. He times how long each one takes to melt.

What changed? Where the ice cube is put (sunny or shady place).

What was measured? How long each ice cube takes to melt.

One thing kept the same: The size of the ice cubes, or the kind of plate.

- ③ Sam grows three bean plants in the same kind of pot on the same windowsill. Every day one plant gets one spoonful of water, one gets two and one gets three. After two weeks he measures how tall each plant is.

What changed? How much water each plant gets each day.

What was measured? How tall each plant grows.

One thing kept the same: The kind of pot, the windowsill, or the time (two weeks).

- ④ Ella puts one spoonful of sugar in a glass of cold water and one spoonful in a glass of warm water. Both glasses hold the same amount of water. She stirs each one at the same speed until she cannot see any sugar, and times how long it takes.

What changed? How warm the water is (cold or warm).

What was measured? How long the sugar takes to dissolve.

One thing kept the same: The amount of sugar, the amount of water, or the stirring.

- ⑤ Ava folds two paper planes from the same kind of paper. One has short wings and one has long wings. She throws each plane the same way and measures how far it flies.

What changed? The size of the wings (short or long).

What was measured? How far each plane flies.

One thing kept the same: The kind of paper, or the way she throws it.

Identifying Variables — Set A — Answer Key

Level 2 · Ages 9-11 · Answers in bold. Accept any answer that means the same.

Independent variable — the one thing you change on purpose.

Dependent variable — what you measure to see what happened.

Controlled variables — things you keep the same so the test is fair.

- ① Priya lets a toy car roll down a ramp set at heights of 10 cm, 20 cm and 30 cm. She measures how far the car rolls across the floor. She uses the same car and the same ramp every time.

Independent variable: The height of the ramp (10, 20 and 30 cm).

Dependent variable: How far the car rolls across the floor (cm).

Two controlled variables: The same car and the same ramp. (Also accept: the same floor.)

- ② Jack ties a metal washer to a string and lets it swing. He tries strings 20 cm, 40 cm and 60 cm long with the same washer, pulling it back the same distance each time. He counts the swings in 30 seconds.

Independent variable: The length of the string (20, 40 and 60 cm).

Dependent variable: The number of swings in 30 seconds.

Two controlled variables: Any two: the same washer; pulled back the same distance; 30 seconds.

- ③ An adult pours 200 mL of warm water at 45 °C into each of three identical cups. Omar wraps one cup in foil, one in wool and one in newspaper. After 20 minutes he measures the temperature of the water in each cup.

Independent variable: The wrapping material (foil, wool or newspaper).

Dependent variable: Temperature after 20 minutes, °C (or the drop in temperature).

Two controlled variables: Any two: 200 mL of water; starting at 45 °C; identical cups; 20 minutes.

Identifying Variables — Set B — Answer Key

Level 2 · Ages 9-11 · Answers in bold. Accept any answer that means the same.

Independent variable — the one thing you change on purpose.

Dependent variable — what you measure to see what happened.

Controlled variables — things you keep the same so the test is fair.

- ① Lucy puts 1, 2, 3 and then 4 sheets of paper between a magnet and a pile of paper clips. Each time she counts how many clips the magnet picks up. She uses the same magnet and the same paper.

Independent variable: The number of sheets of paper (1-4); more sheets = more distance.

Dependent variable: The number of paper clips the magnet picks up.

Two controlled variables: The same magnet and the same kind of paper. (Also accept: the same paper clips.)

- ② Ben adds one teaspoon of sugar to 100 mL of water at 20 °C, 30 °C and 40 °C, stirring steadily the whole time. He times how long the sugar takes to dissolve.

Independent variable: The temperature of the water (20, 30 and 40 °C).

Dependent variable: The time the sugar takes to dissolve (seconds).

Two controlled variables: Any two: one teaspoon of sugar; 100 mL of water; the same steady stirring.

- ③ Zara puts three identical bean plants in a bright window, a shady corner and a dark cupboard. Each plant gets 50 mL of water every day. After two weeks she counts the leaves on each plant.

Independent variable: The amount of light (where each plant is placed).

Dependent variable: The number of leaves after two weeks.

Two controlled variables: Any two: 50 mL of water a day; identical plants; two weeks.

Identifying Variables — Set A — Answer Key

Level 3 · Ages 11-14 · Answers in bold. Accept any answer that means the same.

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① Twenty bean seedlings grow in identical pots of the same potting soil on one windowsill, each given 100 mL of water a day. Each week, groups of five get 0, 5, 10 or 15 drops of liquid fertilizer. After four weeks, every plant's height is measured.

Independent variable (and values):

Drops of fertilizer per week: 0, 5, 10 and 15.

Dependent variable (and units): **Height after four weeks, cm (or mean height per group).**

Three controlled variables: **Any three: identical pots; same potting soil; 100 mL of water a day; same windowsill (light and temperature); four weeks.**

Control group: **The five plants given no fertilizer (0 drops).**

- ② Students take their pulse for 15 seconds after 2 minutes each of sitting, walking and jogging, resting until it is back to the sitting value in between. Each count $\times 4$ gives beats per minute. (Anyone who should not jog can record.)

Independent variable (and values):

Type of activity: sitting, walking, jogging.

Dependent variable (and units): **Heart rate (beats per minute).**

Three controlled variables: **Any three: 2 minutes per activity; 15-second count $\times 4$; rest back to the sitting pulse; the same students.**

Control group: **None — no separate group (accept: sitting as the baseline).**

- ③ Twenty radish seeds are placed on damp paper towel in each of four identical dishes with lids, kept at 5 °C, 15 °C, 25 °C and 35 °C. Each dish gets 10 mL of water. After five days the seeds that have sprouted are counted.

Independent variable (and values):

Temperature: 5, 15, 25 and 35 °C.

Dependent variable (and units): **Number of seeds sprouted (out of 20) after five days.**

Three controlled variables: **Any three: 20 seeds per dish; same type of seed; 10 mL of water; same paper towel; identical dishes with lids; five days.**

Control group: **None (accept a justified baseline).**

Identifying Variables — Set B — Answer Key

Level 3 · Ages 11-14 · Answers in bold. Accept any answer that means the same.

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① A student drops one fizzing vitamin tablet into 200 mL of water at 10 °C, 25 °C and 40 °C, using a new tablet of the same kind each time and never stirring. She times how long each takes to stop fizzing. (Do not drink the solution.)

Independent variable (and values):

Water temperature: 10, 25 and 40 °C.

Dependent variable (and units): **Time to stop fizzing, s (accept: time to dissolve).**

Three controlled variables: **200 mL of water; same kind of tablet; no stirring.**

Control group: **None (accept a justified baseline).**

- ② Parachutes made from the same paper, with canopy areas of 100, 200, 300 and 400 cm², each carry the same paper-clip load. Each is dropped indoors from 2 m, three times, and the mean time to reach the floor is recorded.

Independent variable (and values):

Canopy area: 100, 200, 300 and 400 cm².

Dependent variable (and units): **Mean time to reach the floor (s).**

Three controlled variables: **Any three: drop height 2 m; same load; same paper; indoors (no wind).**

Control group: **None (accept a justified baseline).**

- ③ A piece of pondweed in water is lit by a lamp placed 10, 20, 30 and 40 cm away. The water is kept at 25 °C. At each distance the gas bubbles (which contain oxygen) given off in one minute are counted.

Independent variable (and values):

Distance of the lamp: 10, 20, 30 and 40 cm (light intensity).

Dependent variable (and units): **Number of bubbles given off per minute.**

Three controlled variables: **Any three: same piece of pondweed; water at 25 °C; same lamp; one-minute count.**

Control group: **None (accept: pondweed kept in the dark, as a baseline).**

Spot the Unfair Test — Set A — Answer Key

Level 4 · Ages 10-13 · Answers in bold. Accept any answer that means the same.

- ① Kai wants to know which of two kinds of paper towel soaks up more water. He uses a whole sheet of kind A and half a sheet of kind B, dips each in water for 10 seconds, then squeezes out the water and measures it.

What is the test trying to find out?

Whether the kind of paper towel affects how much water it soaks up.

What is wrong with it? **The towel size changed too: kind B had half as much paper, so it would hold less whatever kind it was. (Also accept: tested once.)**

How would you fix it? **Use pieces of exactly the same size of each kind, and repeat each test a few times.**

- ② Mia wants to see if music helps plants grow. She puts one plant with music playing in her sunny bedroom, and one plant with no music in a dark kitchen.

What is the test trying to find out?

Whether music affects how much a plant grows.

What is wrong with it? **Two things changed at once: the music AND the place (light and temperature). One plant per group is also too few.**

How would you fix it? **Keep all the plants in the same place and conditions, with only the music different. Use several plants in each group.**

- ③ Tom wants to see if the height a ball is dropped from changes how high it bounces. He drops a tennis ball from 50 cm and a basketball from 100 cm.

What is the test trying to find out?

Whether drop height affects how high a ball bounces.

What is wrong with it? **He used two different balls, so the type of ball changed as well as the height. (Also accept: tested once.)**

How would you fix it? **Drop the same ball from each height, test more heights, and repeat each drop.**

Spot the Unfair Test — Set B — Answer Key

Level 4 · Ages 10-13 · Answers in bold. Accept any answer that means the same.

- ① Aisha wants to find out whether sugar or salt dissolves faster. She stirs a spoonful of sugar into hot water and a spoonful of salt into cold water, and times each one.

What is the test trying to find out?

Whether the type of solid (sugar or salt) affects how fast it dissolves.

What is wrong with it? **The water temperature changed too, so she cannot tell which one made the difference. (Also accept: tested once.)**

How would you fix it? **Use the same volume of water at the same temperature, the same mass of each solid, and the same stirring.**

- ② Leo wants to know if heavier toy cars roll farther. He rolls a light car and a heavy car down the same ramp once each and measures how far each one goes.

What is the test trying to find out?

Whether the mass of a car affects how far it rolls.

What is wrong with it? **Each car was tested only once, so one odd result could mislead. The cars may also differ in more than mass (wheels, shape).**

How would you fix it? **Repeat each run several times and use the mean. Better: use one car and add masses to it.**

- ③ A class gives a new plant food to 10 bean plants. After three weeks the plants have grown 12 cm on average, so the class decides that the plant food works.

What is the test trying to find out?

Whether the plant food makes bean plants grow taller.

What is wrong with it? **There is no control group. Without plants that got no plant food, they cannot tell if the plants would have grown 12 cm anyway.**

How would you fix it? **Grow an identical group of 10 plants with no plant food in the same conditions, and compare the two groups.**

Types of Variables — Answer Key

Ages 10-14 · Answers in bold. Accept any answer that means the same.

Independent variable (I) — the one factor you change on purpose.

Dependent variable (D) — the factor you measure to see the effect.

Controlled variable (C) — a factor you keep the same in every trial, so the test is fair.

Control group (G) — a separate group that gets no treatment, used as a baseline to compare against.

Part A — Sort the factors

Experiment: Does plant food make bean plants grow taller? Four groups of bean plants get 0 g, 2 g, 4 g or 6 g of plant food each week. Everything else is kept the same for all the plants. After three weeks, every plant's height is measured.

Write I, D, C or G in each box.

- I** 1. The amount of plant food each week
- D** 2. The height of the plants after three weeks
- C** 3. The amount of water each plant gets
- C** 4. The type of bean plant
- G** 5. The group given 0 g of plant food
- C** 6. The size of the pots
- C** 7. How much light the plants get
- C** 8. How long the test lasts (three weeks)

Note: the 0 g group is also one value of the independent variable.

Part B — Find the variables in each hypothesis

1. If the ramp is higher, then the toy car will roll farther.

Independent: height of the ramp **Dependent:** distance the car rolls

2. If a plant gets more hours of light each day, then it will grow more leaves.

Independent: hours of light each day **Dependent:** number of leaves

3. The warmer the water, the faster salt will dissolve in it.

Independent: temperature of the water **Dependent:** time to dissolve (or how fast)

4. If you exercise for longer, then your heart rate will be higher.

Independent: time spent exercising **Dependent:** heart rate (beats per minute)

Part C — Controlled variable or control group?

Circle the right answer for each statement.

1. Every pot gets exactly the same amount of water.

Controlled variable Control group

2. Some plants get no plant food, so the others can be compared with them.

Controlled variable **Control group**

3. All the plants are kept in the same room.

Controlled variable Control group

Variables in Tables and Graphs — Answer Key

Ages 11-14 · Answers in bold. Accept any answer that means the same.

Example data, made up for practice.

Table 1 (example data)

Ramp height (cm)	Distance rolled (cm)
10	45
20	88
30	131
40	170

Independent variable: Ramp height (cm)

Dependent variable: Distance rolled (cm)

Pattern: As the ramp height increases, the distance rolled increases.

Table 2 (example data)

Water temperature (°C)	Time to dissolve (s)
20	95
30	70
40	52
50	38

Independent variable: Water temperature (°C)

Dependent variable: Time to dissolve (s)

Pattern: As the temperature increases, the time to dissolve decreases.

Table 3 (example data)

Lamp distance (cm)	Bubbles per minute
10	48
20	30
30	18
40	11

Independent variable: Lamp distance (cm)

Dependent variable: Bubbles per minute

Pattern: As the lamp moves farther away, the bubbles per minute decrease.

Plot Table 1 on the graph and label both axes.

y-axis: Distance rolled (cm)

